Mount-In-Propagation Formulation Revision

sensor-fusion documentation

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1 Scope

This document records the formulation change that moved mount alignment into the inertial propagation model for both the runtime ESKF and the loose reference filter. It is intentionally a before/after document. The full steady-state formulations remain in the dedicated ESKF and loose formulation PDFs, while this document explains why the revision was needed, what changed mathematically, and how the implementation maps to the revised equations.

The motivating failure was observed on a synthetic figure-eight profile with explicit vehicle roll excitation. With truth-quality inputs, both filters previously converged to a near-correct composed vehicle attitude while retaining a persistent mount roll error. The old formulation allowed vehicle roll error and residual mount roll error to trade against each other. The revised formulation removes that practical gauge by making mount errors affect strapdown attitude and velocity propagation.

2 Notation and Frames

Symbol	Meaning
b	Raw IMU body frame.
v	Physical vehicle frame, forward-right-down.
n	Local NED navigation frame used by the ESKF.
e	ECEF frame used by the loose reference filter.
s	Old seeded frame produced by pre-rotating raw IMU with a coarse mount seed.
c	Old corrected vehicle frame after applying residual mount.
q_{vb}	Physical vehicle-to-body mount quaternion, with $x_b = C_{bv}x_v$.
q_{nv}	Vehicle-to-NED attitude quaternion, with $x_n = C_{nv}x_v$.
q_{ev}	Vehicle-to-ECEF attitude quaternion, with $x_e = C_{ev}x_v$.

Rotations follow the project convention

$$x_a = C_{ab}x_b, \qquad C(q_1 \otimes q_2) = C(q_1)C(q_2). \tag{1}$$

Small-angle perturbations use the first-order quaternion

$$\delta q(\delta\theta) \approx \begin{bmatrix} 1 & \frac{1}{2}\delta\theta^T \end{bmatrix}^T. \tag{2}$$

3 Old Formulation

3.1 State Split

The old ESKF nominal state was

$$x_{\text{old}} = \begin{bmatrix} q_{ns} & v_n^T & p_n^T & b_g^T & b_a^T & q_{cs} \end{bmatrix}^T, \tag{3}$$

where q_{ns} rotated the seeded frame into NED and q_{cs} rotated the seeded frame into the corrected vehicle frame. The corresponding error state was

$$\delta x_{\text{old}} = \begin{bmatrix} \delta\theta_s & \delta v_n & \delta p_n & \delta b_g & \delta b_a & \delta\psi_{cs} \end{bmatrix}^T. \tag{4}$$

The loose filter had the same structure in ECEF:

$$x_{\text{old,loose}} = \begin{bmatrix} q_{es} & v_e^T & p_e^T & b_g^T & b_a^T & s_g^T & s_a^T & q_{cs} \end{bmatrix}^T. \tag{5}$$

3.2 IMU Pre-Rotation

The align filter provided a coarse physical mount seed q_{vb}^0. Before the ESKF or loose propagation step, the runtime converted raw IMU samples into the seeded frame:

$$\Delta\theta_s = C_{bv}^{0^T}\Delta\theta_b, \tag{6}$$

$$\Delta v_s = C_{bv}^{0^T}\Delta v_b. \tag{7}$$

The filters then propagated using $\Delta\theta_s$ and Δv_s. The residual mount q_{cs} was held constant during propagation:

$$q_{cs}^+ = q_{cs}^-. \tag{8}$$

3.3 Old ESKF Propagation

The old ESKF prediction was effectively

$$\Delta\theta_s^c = \Delta\theta_s - b_g\Delta t, \tag{9}$$

$$\Delta v_s^c = \Delta v_s - b_a\Delta t, \tag{10}$$

$$q_{ns}^+ = \text{normalize}\left(q_{ns}^- \otimes \delta q(\Delta\theta_s^c)\right), \tag{11}$$

$$v_n^+ = v_n^- + C_{ns}\Delta v_s^c + g_n\Delta t, \tag{12}$$

$$p_n^+ = p_n^- + v_n^-\Delta t. \tag{13}$$

The important structural point is that q_{cs} did not appear in these equations. Therefore the generated propagation Jacobian had no dynamic sensitivity from mount error into attitude or velocity:

$$\frac{\partial q_{ns}^+}{\partial\delta\psi_{cs}} \approx 0, \qquad \frac{\partial v_n^+}{\partial\delta\psi_{cs}} \approx 0. \tag{14}$$

3.4 Old NHC Measurement

The old non-holonomic constraint used the corrected vehicle velocity

$$v_c = C_{cs}C_{ns}^T v_n. \tag{15}$$

The lateral and vertical constraints were

$$z_y = e_y^T v_c = 0, \qquad z_z = e_z^T v_c = 0. \tag{16}$$

This gave a direct residual-mount measurement Jacobian:

$$H_{\psi_{cs}} = \frac{\partial(e_i^T C_{cs}C_{ns}^T v_n)}{\partial\delta\psi_{cs}}. \tag{17}$$

3.5 Why This Was Insufficient

The old model could produce a correct composed vehicle-frame quantity while splitting the error incorrectly:

$$C_{cv} = C_{cs}C_{sn} \tag{18}$$

could remain nearly correct even if the attitude and residual mount separately were wrong. In the observed synthetic roll-excitation case, the filters settled to approximately

$$\text{mount roll error} \approx +0.45^\circ, \tag{19}$$

$$\text{vehicle roll error} \approx -0.45^\circ, \tag{20}$$

with a near-zero sum. Since q_{cs} did not affect propagation, the IMU dynamics did not provide an independent signal to decide whether the correction belonged in attitude or mount. Tuning covariance could change allocation, but it could not remove the structural ambiguity.

4 Revised Formulation

4.1 State Meaning

The revised ESKF keeps the same state dimension and Rust field layout, but changes the meaning of the quaternion formerly named `qcs`:

$$q_{cs} \quad \text{is now interpreted as} \quad q_{vb}. \tag{21}$$

That is, `qcs0..qcs3` store the physical vehicle-to-body mount, not a residual seeded-frame-to-vehicle correction.

The revised ESKF nominal state is

$$x_{\text{new}} = \begin{bmatrix} q_{nv} & v_n^T & p_n^T & b_g^T & b_a^T & q_{vb} \end{bmatrix}^T, \tag{22}$$

and the revised error state is

$$\delta x_{\text{new}} = \begin{bmatrix} \delta\theta_v & \delta v_n & \delta p_n & \delta b_g & \delta b_a & \delta\psi_{vb} \end{bmatrix}^T. \tag{23}$$

The loose filter analog is

$$x_{\text{new,loose}} = \begin{bmatrix} q_{ev} & v_e^T & p_e^T & b_g^T & b_a^T & s_g^T & s_a^T & q_{vb} \end{bmatrix}^T. \tag{24}$$

4.2 Raw IMU Propagation Through Mount

The filters now consume raw body-frame IMU increments. Bias-corrected body-frame increments are

$$\Delta\theta_b^c = \Delta\theta_b - b_g\Delta t, \tag{25}$$

$$\Delta v_b^c = \Delta v_b - b_a\Delta t. \tag{26}$$

The current mount rotates them into the vehicle frame:

$$\Delta\theta_v^c = C_{bv}(q_{vb})^T\Delta\theta_b^c, \tag{27}$$

$$\Delta v_v^c = C_{bv}(q_{vb})^T\Delta v_b^c. \tag{28}$$

The revised ESKF propagation is

$$q_{nv}^+ = \text{normalize}\left(q_{nv}^- \otimes \delta q(\Delta\theta_v^c)\right), \tag{29}$$

$$v_n^+ = v_n^- + C_{nv}\Delta v_v^c + g_n\Delta t, \tag{30}$$

$$p_n^+ = p_n^- + v_n^-\Delta t, \tag{31}$$

$$q_{vb}^+ = q_{vb}^-. \tag{32}$$

For loose, the same structure is used with ECEF gravity and Earth rotation terms:

$$\omega_v = C_{bv}(q_{vb})^T \omega_b^c, \tag{33}$$
$$f_v = C_{bv}(q_{vb})^T f_b^c, \tag{34}$$
$$\dot{q}_{ev} = \frac{1}{2} q_{ev} \otimes \begin{bmatrix} 0 & \omega_v^T \end{bmatrix}^T + \text{Earth-rate correction}, \tag{35}$$
$$\dot{v}_e = C_{ev} f_v + g_e(p_e) - 2\omega_{ie,e} \times v_e, \tag{36}$$
$$\dot{p}_e = v_e. \tag{37}$$

4.3 New Propagation Sensitivities

The essential new Jacobian blocks are the mount sensitivities in propagation. For a small mount perturbation,

$$C_{bv}(\delta q(\delta\psi_{vb}) \otimes q_{vb})^T x_b \approx C_{bv}(q_{vb})^T x_b - C_{bv}(q_{vb})^T \left[x_b\right]_\times \delta\psi_{vb}. \tag{38}$$

Therefore, to first order,

$$\frac{\partial \Delta\theta_v^c}{\partial \delta\psi_{vb}} \approx -C_{bv}^T \left[\Delta\theta_b^c\right]_\times, \tag{39}$$
$$\frac{\partial \Delta v_v^c}{\partial \delta\psi_{vb}} \approx -C_{bv}^T \left[\Delta v_b^c\right]_\times. \tag{40}$$

These terms create covariance between mount error, attitude error, and velocity error during propagation. GNSS position/velocity and NHC can then correct mount through this propagated cross-covariance. This is the observability path that was missing before.

4.4 Revised NHC Measurement

Since q_{nv} now represents the physical vehicle attitude, NHC no longer needs the mount quaternion directly:

$$v_v = C_{nv}^T v_n. \tag{41}$$

The ESKF NHC rows are

$$z_y = e_y^T C_{nv}^T v_n = 0, \qquad z_z = e_z^T C_{nv}^T v_n = 0. \tag{42}$$

The loose rows are

$$z_y = e_y^T C_{ev}^T v_e = 0, \qquad z_z = e_z^T C_{ev}^T v_e = 0. \tag{43}$$

Thus

$$\frac{\partial z_i}{\partial \delta\psi_{vb}} = 0 \tag{44}$$

for the instantaneous NHC row. This is expected and desired. Mount is no longer directly nudged by NHC in a single update; it is corrected through the state correlations generated by propagation.

5 Before/After Comparison

Aspect	Before	After

Mount state meaning	Residual seeded-frame-to-vehicle q_{cs}.	Physical vehicle-to-body q_{vb}, stored in the same Rust fields.
IMU input to propagation	Raw body IMU was pre-rotated by align seed before entering filter.	Raw body IMU enters filter; generated propagation rotates through current q_{vb}.
Mount in propagation	No direct appearance in attitude or velocity propagation.	Directly affects attitude and velocity propagation through C_{bv}^T.
NHC measurement	$v_c = C_{cs}C_{ns}^T v_n$, direct residual-mount Jacobian.	$v_v = C_{nv}^T v_n$, no direct mount Jacobian.
Mount observability path	Mostly NHC direct row plus covariance artifacts.	Propagation cross-covariance plus GNSS/NHC corrections.
Failure mode	Vehicle attitude and residual mount could absorb equal/opposite errors.	Wrong mount causes wrong inertial propagation, so residuals push mount.

6 Implementation Mapping

6.1 Generated ESKF

The symbolic source is `ekf/eskf.py`. The generated Rust fragments are included by `ekf/src/generated_eskf.rs`. The important changes are:

- `propagate_nominal(...)` now takes `qcs` as q_{vb}.
- Nominal prediction rotates $\Delta\theta_b$ and Δv_b through C_{bv}^T.
- Error transition and noise input include mount-to-attitude and mount-to-velocity propagation sensitivity.
- Body velocity observations no longer depend on `qcs`.

6.2 Runtime ESKF Facade

`ekf/src/fusion.rs` no longer pre-rotates IMU before calling `RustEskf::predict`. It initializes `nominal.qcs*` from the align or reference mount seed. In follow-align mode, the facade writes the current align mount into `nominal.qcs*` and keeps mount error states frozen.

6.3 Loose Reference Filter

The symbolic source is `ekf/ins_gnss_loose.py`. The generated Rust fragments are included by `ekf/src/generated_loose.rs`. The important changes are:

- The transition matrix uses `qcs` as q_{vb} in the IMU rotation terms.
- Loose nominal propagation in `ekf/src/loose.rs` rotates raw body-frame gyro and accelerometer measurements into the vehicle frame.
- Loose NHC rows observe $C_{ev}^T v_e$ and have no direct mount support.
- The visualizer seeds loose `qcs` with align mount and plots it directly as physical mount.

7 Covariance and Initialization Implications

Moving mount into propagation makes mount observable through cross-covariance, so the initial covariance must allow the filter to adjust the attitude/mount split. This matters most for the loose filter because it initializes vehicle attitude from a yaw-only state. If the loose vehicle-attitude covariance is too small, the filter remains overconfident in an attitude that cannot yet be known well enough, and the mount state cannot fully absorb the coarse seed error.

The visualizer default was therefore changed from

$$\sigma_{\theta,\text{loose}} = 2^\circ \tag{45}$$

to

$$\sigma_{\theta,\text{loose}} = 20^\circ. \tag{46}$$

This is not an ad-hoc correction to one maneuver. It reflects the weaker initial information content of the loose yaw-only attitude seed.

8 Synthetic Validation

On the truth-input figure-eight roll-excitation profile without GNSS fault, the old filters settled at nonzero mount roll error:

Filter	Old final mount qerr	New final mount qerr
ESKF	approximately 0.46°	0.0277°
Loose	approximately 0.45°	0.0321°

With an early GNSS velocity fault of $+0.5\,\text{m/s}$ north over $0 \ldots 360\,\text{s}$, the revised formulation still converges:

Filter	Final mount qerr
ESKF	0.0887°
Loose	0.0545°

In that same faulted case, the loose GNSS position gate accepted all GNSS position updates after the revision. Before this change, the bad state split could drive the loose prediction far enough away that the full 3D position gate rejected valid measurements for the rest of the run.

9 Test Coverage

The following tests exercise the revised model:

- `ekf/tests/eskf_nhc_jacobian.rs`: finite-difference check for the direct vehicle-frame ESKF NHC Jacobians.
- `ekf/tests/nhc_eskf_loose_equivalence.rs`: verifies ESKF and loose NHC rows and updates remain equivalent after the attitude basis transform.
- `ekf/tests/loose_mount_observability.rs`: verifies NHC does not directly move mount under the new formulation; mount is corrected through propagation correlations.
- The synthetic GNSS/INS test verifies that roll-excitation now makes internal mount estimates converge.

10 Remaining Follow-Up

The existing ESKF and loose formulation PDFs still describe portions of the old seeded-frame terminology. They should be updated to use this document as the source of truth for mount semantics, or merged into a single current formulation reference after the revised behavior is accepted on the full experimental data suite.